

Cogs 105 Final Project

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Data Explanation

The dataset included a wide range of variables, but several were removed because they did not contribute meaningful predictive information. ID was dropped because it is simply an identifier and does not relate to ASD status. Ethnicity and country_of_res were also removed because many categories had only a few individuals, making them unreliable and not predictive in the model. The result variable was excluded as well because it did not match the individual A1–A10 scores, which raised concerns about its accuracy. Instead, a new variable was created, AQ_sum, which represents the total score across A1–A10 and more accurately reflects the intended Autism Quotient measure.

The AQ Test consists of 10 items commonly used as a screening tool for autism. Each question offers four response options—*Definitely agree*, *Slightly agree*, *Slightly disagree*, and *Definitely disagree*—but the scoring simplifies these into binary agree/disagree categories. Scores for the AQ_sum variable range from 0–10, and a total of 6 or higher indicates an elevated likelihood of autism, while scores below 6 suggest a lower likelihood.

Variables

ID: ID of patient	Autism: whether an immediate family member has been diagnosed with autism
A1_Score to A10_Score: Score based on Autism Spectrum Quotient. 10-item screening	Country_of_res: country of residence of the patient
Age: age of the patient in years	Used_app_before: whether the patient has undergone screening before
Gender: gender of the patient	Result: Score for AQ1-10 testing
Ethnicity: ethnicity of the patient	Age_desc: whether the patient is over or under 18
Jaundice: whether the patient had jaundice at the time of birth	Relation: relation of the patient who completed the test
AQ_sum: added variable adding together the scores from the AQ1-10	Class/ASD: was the patient diagnosed with ASD. 0 for no, and 1 for yes. Target column

Objective of the Analysis

This analysis asked three questions:

1. Which features best predict an ASD diagnosis?
2. Can the individual AQ-10 items predict ASD better than using all of the predictors?
3. Do early in life or demographic factors influence the likelihood of testing positive for ASD?

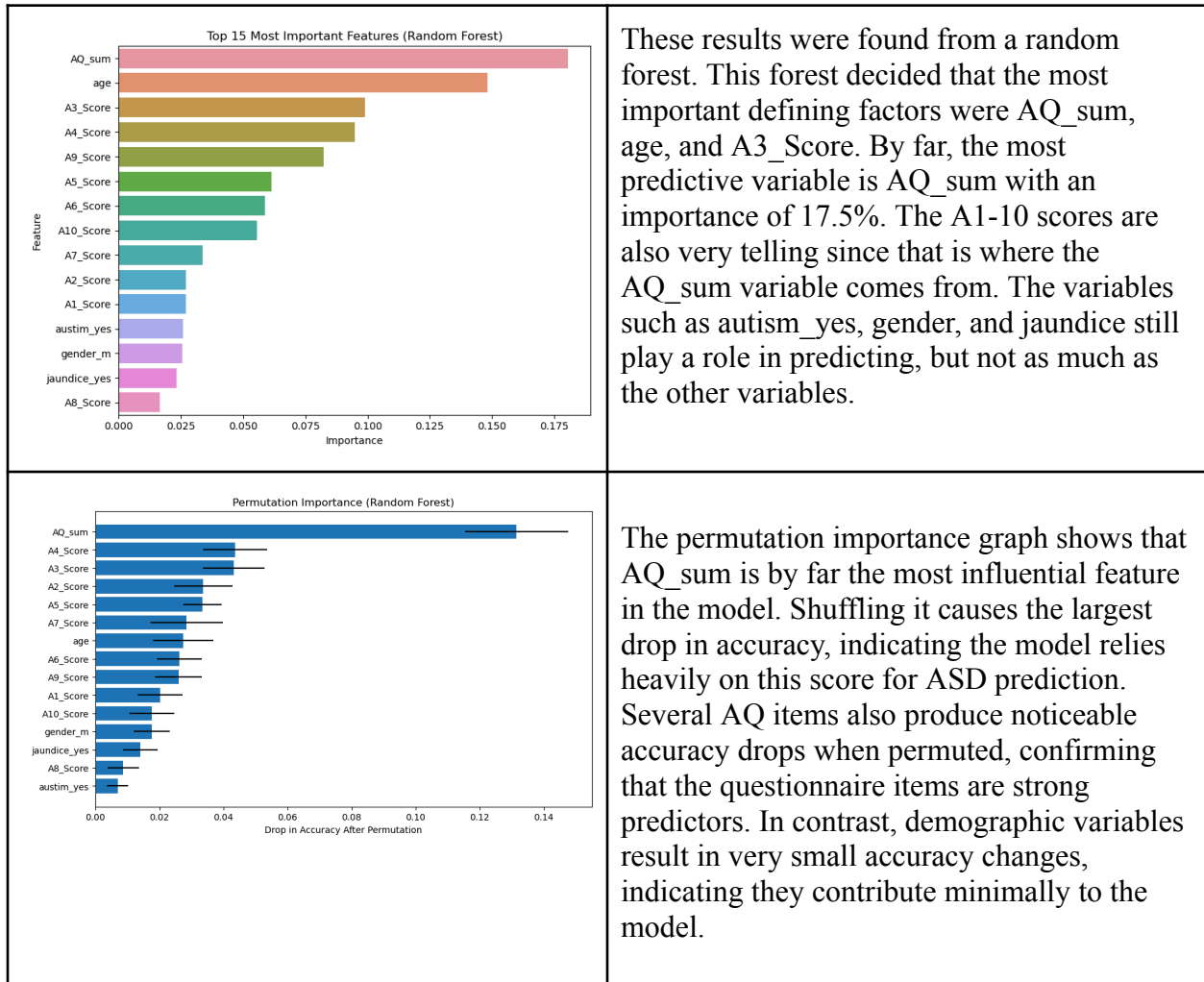
The primary objective of this analysis is to evaluate the factors that influence the likelihood of receiving an Autism Spectrum Disorder (ASD) diagnosis based on a dataset containing AQ questionnaire responses and demographic information. This involves building and validating a Random Forest classifier to identify which questionnaire items and demographic variables contribute most strongly to ASD classification.

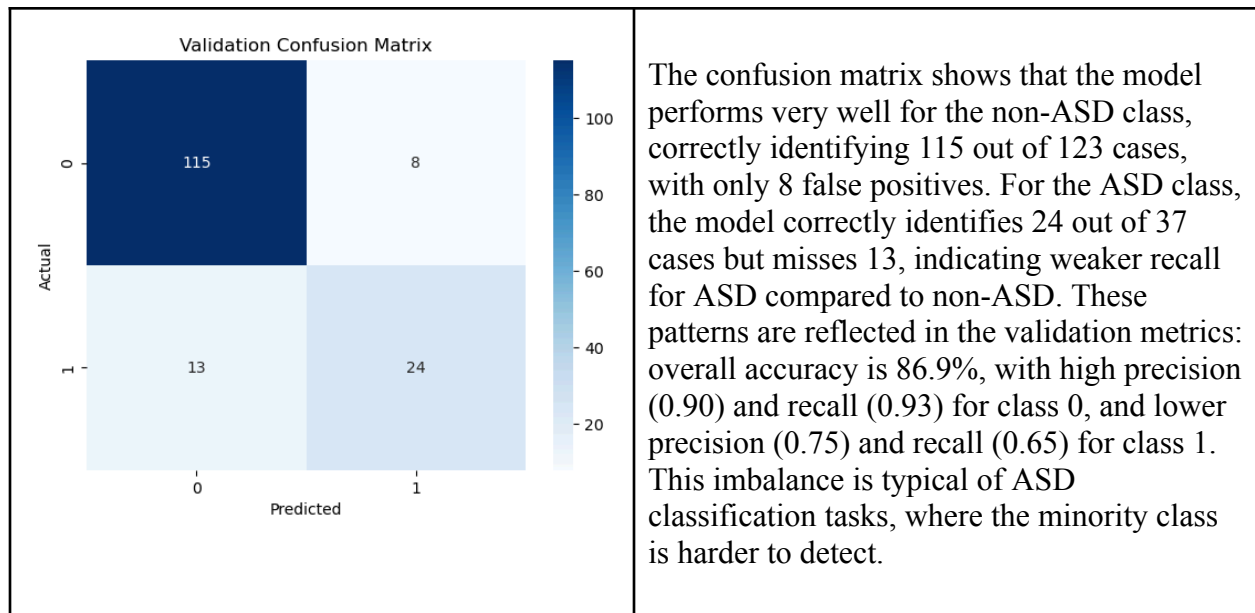
The analysis also examined how well ASD could be predicted using only the AQ-related variables, including the individual questionnaire items (A1–A10) and the total AQ sum score. These features represent the core components of the autism screening tool and directly measure behaviors and traits associated with ASD. When used on their own, the AQ items and AQ_sum captured most of the predictive signal in the dataset, coming close to—but not exceeding—the performance achieved when demographic and early-life variables were included. This indicates that the questionnaire itself is the primary driver of accurate ASD classification, and that the AQ scores provide nearly all of the information needed for reliable prediction, even though adding demographic predictors offers a small additional improvement.

Additionally, the analysis examines whether demographic or early-life factors, such as age group (under 18 vs over 18), meaningfully affect ASD detection. This helps determine whether the screening questionnaire performs consistently across age groups or whether certain demographic subgroups are more likely to screen positive. The broader goal is to assess both the predictive accuracy of the model and the relative importance of the different predictors involved.

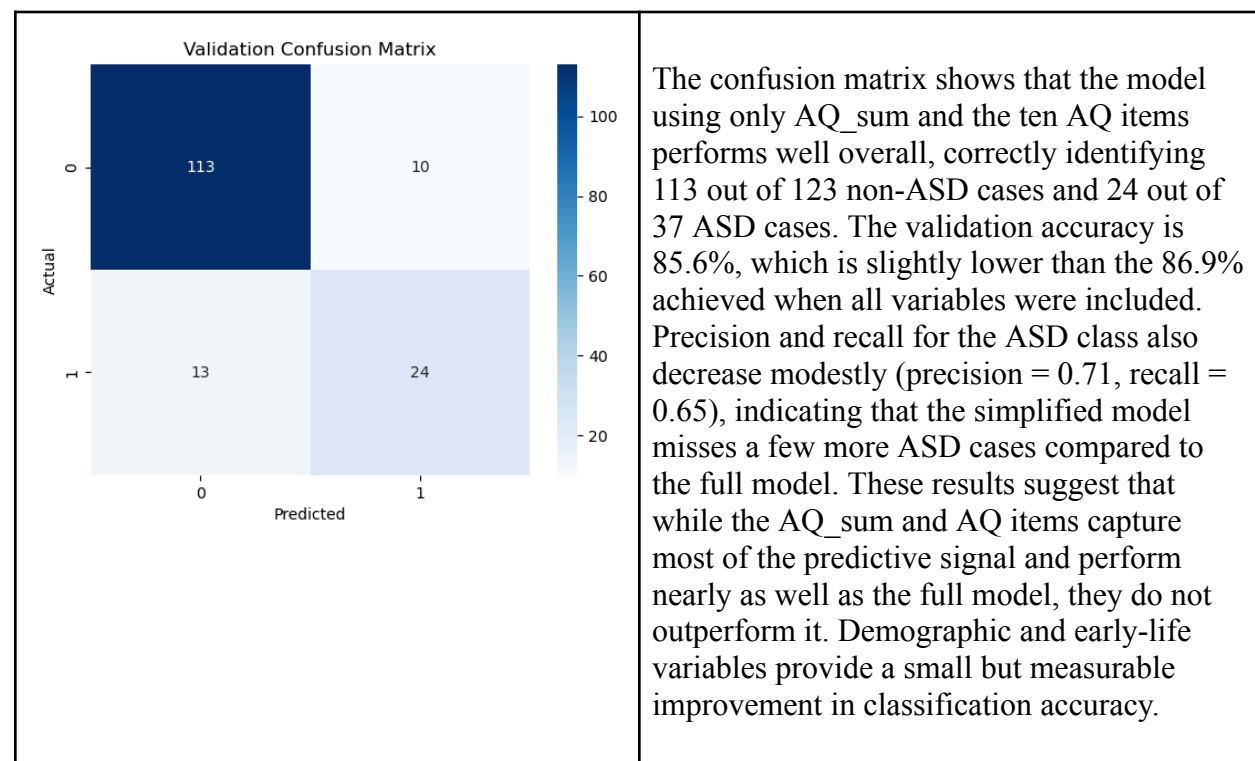
Graphs and Result

Results from Question 1

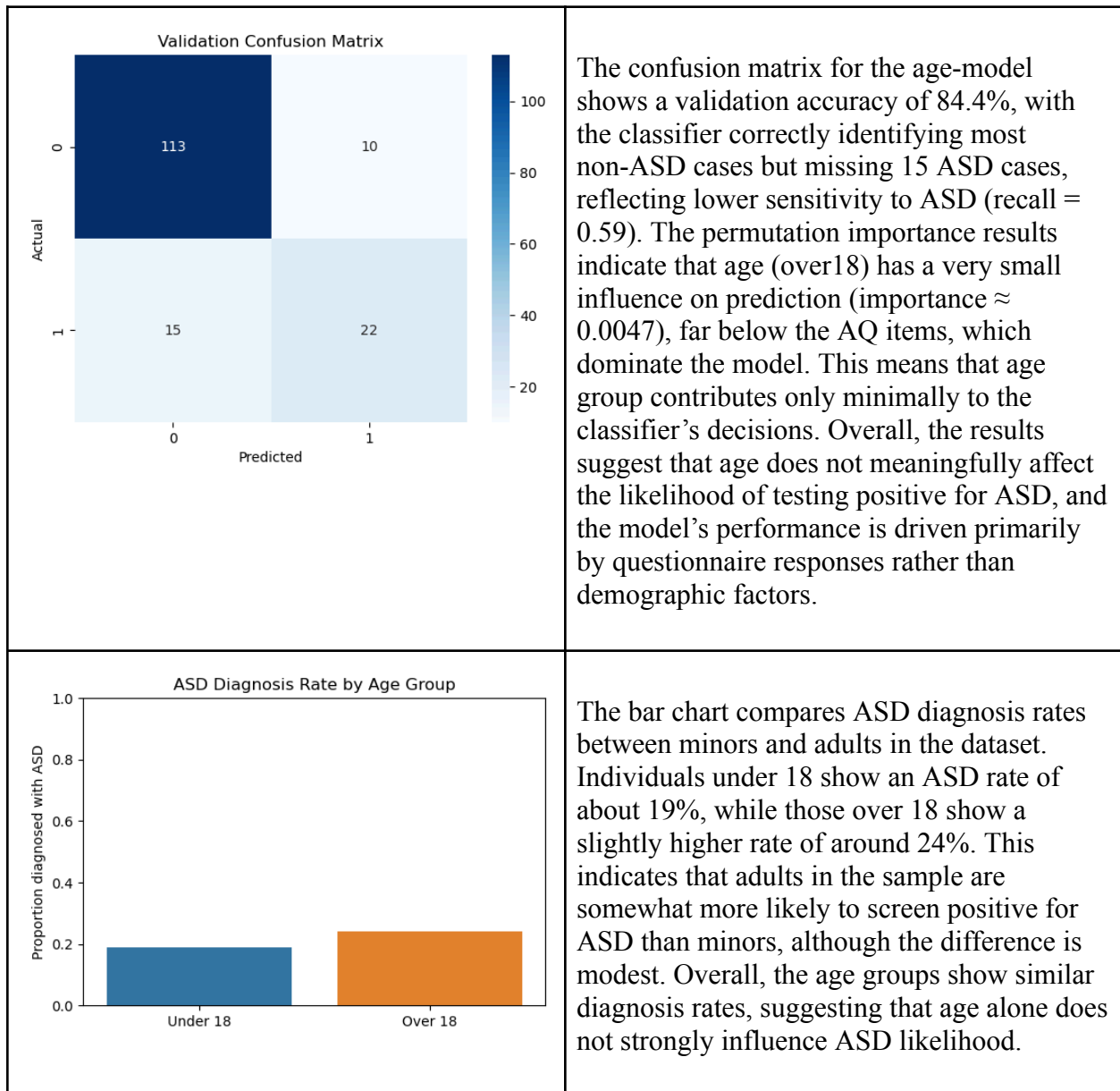




Results from Question 2



Results from Question 3



Discussion

This analysis demonstrated that the AQ questionnaire is the strongest predictor of ASD classification in the dataset. Models using only the AQ items performed nearly as well as models including demographic and early-life variables, indicating that the questionnaire itself captures most of the predictive signal. Although adding demographic features provided a small

improvement in accuracy, permutation importance showed that variables such as age group, gender, jaundice history, and family autism history contributed only minimally to the model. In particular, age (over 18 vs. under 18) had very low predictive value, suggesting that the questionnaire performs similarly across age groups. The diagnostic rates by age group showed only modest differences, further supporting the conclusion that demographic factors do not meaningfully influence ASD prediction in this dataset.

Although the model performed well overall, it was less effective at correctly identifying ASD cases compared to non-ASD cases. This suggests that additional information not included in the dataset might help improve detection of ASD. For example, having more detailed behavioral or developmental data, information about social functioning, or a larger sample of ASD cases could strengthen the model's ability to distinguish between groups. It may also be useful to examine whether the AQ questions measure symptoms similarly across different age groups or other subgroups. Collecting more balanced and diverse data would help confirm whether the patterns observed here generalize beyond this sample.

Overall, the analysis shows that the AQ questionnaire is an effective predictor of ASD, but further data and more detailed demographic information would be valuable for improving model performance and deepening our understanding of ASD screening patterns.